TEACHING STATEMENT

CHRISTOPHER HERBIG

Mathematics is a subject that I enjoy learning and teaching for the opportunities it provides us to use creative (and sometimes beautiful) forms of reasoning. Regardless of the setting in which I find myself teaching, I seek to emphasize this creativity beauty rather than simply emphasizing its utility. I approach teaching by aiming to inspire curiosity, making the underlying intuition and motivation behind every concept explicit. By doing so, I strive to make the subject accessible to all students and thereby giving them the insight to appreciate mathematics on a deeper level.

As an instructor at Kishwaukee College, I taught introductory mathematics to a highly diverse group of non-STEM majors, many of whom were future nurses, aspiring artists, and returning adult learners. It became clear early on that unbroken lecturing would leave students behind. I shifted to an inquiry-based model similar to the one we implement for certain courses at Northern Illinois University. Here, I introduce a concept briefly, then immediately have students work on concrete problems, either alone or in groups. This allows me to circulate the room and clear up the various idiosyncratic misconceptions that the students may hold. By having them handle the material immediately, I find that they absorb the concepts more readily, and I can adapt to their individual abilities.

When introducing new material, I focus on making the concepts memorable by separating them into motivation, i.e., why the problem exists and is worth approaching, and intuition, i.e., the approach to the problem. Motivation hooks the students. For instance, before teaching the normal distribution, I show real-world data on birth weights and shoe sizes to demonstrate its ubiquity in nature. The intuition then demystifies the mechanics. Carefully walking through the logical progression gives students a sense of insight they can apply to ostensibly unrelated problems. Emphasizing both gives them an understanding that lasts well after the semester ends.

To keep students on track, I rely on regular, low-stakes assessments. Borrowing from the college algebra model at Northern Illinois University, I give weekly quizzes based loosely on upcoming exam questions. This gives students a clear sense of what needs studying without the pressure of a major test. Simultaneously, it allows me to catalog recurring mistakes and address those lapses in understanding during class while the material is still fresh.

Ultimately, my work is an opportunity to help students from all backgrounds appreciate the beauty of mathematics as I do. Regardless of the course level, I aim

to impart a lasting respect for the discipline and the confidence to reason through complex problems logically.

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